Java Assignment 11
Marks: 20

General Instructions

- Attempt all three questions. Each question must be written as a separate, compilable Java program.
- Follow proper naming conventions, add meaningful comments and use methods to keep the code modular.
- Do not use any built-in method that directly computes the Longest Common Subsequence.

Q1. File Handling: Word Statistics and Copy (5 marks)
Write a Java program that takes two file names from the command line: java FileStats input.txt output.txt.

- (a) Use FileReader and BufferedReader to read input.txt line by line. Count the total number of lines, words and characters.
- (b) Use FileWriter and BufferedWriter to write each line to output.txt with a line number in front of it (e.g. 1: Hello world). Add the three counts at the end of the file.
- (c) Check that the input file exists using File.exists(). Handle FileNotFoundException and IOException properly. Print the absolute path of the output file when done.

Q2. Multithreading: Parallel Number Tasks (5 marks)
Write a Java program that creates two threads in different ways:

- (a) EvenPrinter, created by extending Thread, prints the first 10 even numbers
- (b) OddPrinter, created by implementing Runnable, prints the first 10 odd numbers.
- (c) In both, use Thread.sleep(500) between prints and set a custom name with setName(). Each print statement must show the thread name.
- (d) In main(), print the current thread's details with Thread.currentThread(). Start both threads. Print getState() and Thread.activeCount() before and after starting them.

Q3. Networking: TCP Client-Server Chat (10 marks)
Write a TCP client-server program in Java using the java.net package.

- (a) **Server (MyServer.java):** Create a ServerSocket on port 5000 with a queue length of 5. Call accept() to wait for a client. Get DataInputStream and DataOutputStream from the socket.
- (b) **Client (MyClient.java):** Connect to the server with Socket. Send a number n entered by the user.

- (c) The server receives n, computes its factorial and sends the result back. The client prints it.
- (d) Close all streams and sockets properly on both sides. Use InetAddress to print the server's host name and IP address on the client side.